
THE FERROMAGNET IS THE WITNESS OF THE CIRCLE

Lee–Yang Zeros on the Fixed Set of Inversion, the Unit Circle as the Critical Line, Positivity as the Supplier That Seats the Zeros, and the Antiferromagnet That Leaves Them

The Closure Template of the Riemann Hypothesis Read on the Ising Ferromagnet; Fifty-Eight Laws Proved in Core Lean 4 and Nine Checks Executed in Fortran; a Witness of the Hardware Closure Series; Every Physical Claim at Corroboration Grade and No Higher

FOUNDATIONS OF PHYSICS · THE HARDWARE CLOSURE SERIES

Positivity seats the zeros on the circle

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ABSTRACT. In 1952 Lee and Yang proved that the partition function of an Ising ferromagnet, read as a polynomial in the fugacity z of the magnetic field, has every zero on the unit circle. It remains the one mechanism in physics by which a positivity alone forces the zeros of a function onto a locus. This paper reads it on the seat of the author’s closure of the Riemann Hypothesis, the reflection $s \mapsto 1 - \bar{s}$ whose fixed set is the critical line. The logarithmic carrier $s = \frac{1}{2} + \frac{1}{2} \ln z$ sends inversion in the unit circle, $z \mapsto 1/\bar{z}$, to that reflection and the circle to the line: Lee and Yang’s circle is the critical line in another chart. We prove in core Lean 4, fifty-eight laws with no axiom declared and no use of choice, that spin flip keeps every domain wall, so the partition function is palindromic and its zeros pair under inversion; that the four-spin ring, its sixteen states counted, has every zero on the circle exactly when its coupling is ferromagnetic and loses one below the circle’s reach for an antiferromagnet; and that the two-spin system likewise. Symmetry pairs the zeros across the locus; positivity seats them on it. A Fortran twin runs nine checks with zero failures: 300 random ferromagnets on six to ten spins keep every zero on the circle, 300 antiferromagnets each lose at least one, and on two-dimensional tori below the critical temperature the zero nearest the positive axis closes in from 0.412 to 0.269 to 0.202 radians as the lattice grows, the onset of a phase transition read in its zeros. The paper supplies nothing toward the question whether Riemann’s function lies in the Lee–Yang class, which would be a sufficient route to the hypothesis and not an equivalent of it; it decides nothing about the zeros of ζ .

KEYWORDS *Lee–Yang theorem · partition function zeros · Ising model · ferromagnetism · phase transitions · unit circle · critical line · positivity · de Bruijn–Newman · Lean 4*

STATUS: WITNESSED · CROSSED · CONFIRMED on the image · Row 2 (the least-erasure face’s shape: a positivity that seats the zeros); every open row (the barrier, in the antiferromagnet’s pairs) · channel: kinetic · Lean: SPHYS_Lee_Yang . lean · Fortran: LeeYang_Twin . f90

1. THE READER'S FRAME

This paper is a witness of the hardware closure series. The Riemann closure reduced the hypothesis to one bit with five faces. One of them, least erasure, has a physical shape the arrow witness named: a registration that erases nothing because nothing stands off its locus, kept there by one sign. The Lee–Yang theorem is that shape in its sharpest physical form. A ferromagnet's couplings all have one sign, and every zero of its partition function lies on the unit circle.

It claims three things. The unit circle is the critical line under a logarithmic carrier forced by the symmetry. Spin flip, the ferromagnet's own symmetry, pairs the zeros across that locus as the functional equation pairs Riemann's zeros across the line, and seats none of them. Positivity seats them: the ferromagnet keeps every zero on the circle, and the antiferromagnet, with couplings of the other sign, loses them.

It refuses the bridge to Riemann's function. Whether Riemann's Ξ is the Fourier transform of a function in the Lee–Yang class is a question that would give a sufficient route to the hypothesis if answered yes. It is not an equivalent of the hypothesis, and this paper supplies nothing toward it. A reader who arrives from the conventional frame will scan this paper for that bridge, find it typed and not crossed, and be tempted to report the absence. It was not attempted.

Negative warrants. Three, stated so that no reader supplies them.

- No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade.
- No derivation of the hypothesis from the axioms of set theory is claimed: what a sentence over every admissible instance adds to a crossing is exactly the unactuated.
- The general Lee–Yang theorem is imported, and only its smallest instances are proved here; the twin corroborates it on larger systems.

Five terms.

- The *fugacity* is $z = e^{-2\beta h}$ for a magnetic field h at inverse temperature β .
- The *partition function* is a polynomial in z whose coefficient c_m sums the Boltzmann weights of the states with m down spins.
- A coupling is *ferromagnetic* when it favours aligned spins and *antiferromagnetic* when it favours opposed spins.
- A polynomial is *palindromic* when $c_m = c_{N-m}$.
- A *keyed bit* is a yes-or-no property no function of a given record decides.

2. THE SPINE

The Riemann closure has four parts: the **seat**, the critical line as the fixed set of $s \mapsto 1 - \bar{s}$; the **address**, every reading of the value one proposition; the **division**, a certified part joined to an uncertified tail; and the **closure**, registration keeping a zero's height and forgetting its side, the hypothesis holding exactly when it erases nothing. This paper reads the seat on the fugacity plane in Section 3, the pairing symmetry in Section 4, the positivity that seats in Section 5, the sign that leaves in Section 6, the phase transition in Section 7, the measurements in Section 8, and the bridge to Riemann's function, typed and not crossed, in Section 9.

The row-generic theorems carry the template, and the engine proves each on its chart: every point off the line has a partner with one record (`off_locus_pair`); a carrier proves the value on its image, and coverage extends it (`value_on_image`, `kinetic_crossing`).

The witness ledger. The ledger of the series, with this paper's entries.

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
2 · Riemann Hypothesis	crossed on the kinetic channel at the root's grade; the electron named as the seat	the six witnesses; One Actuating Substrate; The Geometric Nature of Light; Electromagnetism on the Riemann Locus; Immanent Gravity; Spin and Statistics; The Weak Force Reads the Orientation; Resonances; The Ferromagnet Is the Witness of the Circle	the seat on every reading so far; the least-erasure face's shape in a ferromagnet: symmetry pairs the zeros, positivity seats them on the circle, the antiferromagnet leaves	WITNESSED · CROSSED · CONFIRMED on the image: the shape of a positivity that seats	the crossing at the root's grade; the image at theorem grade on construction and measured grade on data	none on its channel
3a · Navier–Stokes, unforced	crossed kinetically	The Fluid Is the Witness	the priced branch exact	WITNESSED · CROSSED · CONFIRMED	the root's grade	none on its channel
4 · Yang–Mills	crossed kinetically	The Proton Is the Lock	the colour lock	WITNESSED · CROSSED · CONFIRMED on the image	the root's grade	none on its channel
7 · Poincaré	crossed at full grade	The Arrow Has Two Branches; Immanent Gravity	gravity's fold	CROSSED, WITNESSED from both sides	full grade	none
23a · Pair correlation	crossed kinetically	The Magnet Is the Witness of the Pair Correlation; Electromagnetism on the Riemann Locus; Resonances	the magnetic term; closed and open systems	WITNESSED · CROSSED · CONFIRMED on the corrected carrier	the root's grade	none on its channel
every open row · the barrier	two worlds over one record	the arrow, light, electromagnetism, spin and statistics, weak, resonance papers; The Ferromagnet Is the Witness of the Circle	the antiferromagnet's zero pairs off the circle, each pair one palindromic record	BARRIER WITNESSED; the rows stay pending	theorem and corroboration	one equivariant carrier each
1 · P versus NP	the dot	The Arrow Has Two Branches	Landauer's floor prices every step	THE DOT STANDS	corroboration	none
every witness · the hardware	the carriers built one by one	One Actuating Substrate	both involutions carried onto the fold	WITNESSED: one substrate carries every seat	theorem and corroboration	none

WITNESSED: a constructed witness arrived and was built and run. CROSSED: the row's value holds on its channel's own class at the printed grade. CONFIRMED: the witness's image is compliant as built or measured (value_on_image).

Why this carrier, and not another. The carrier is not chosen. Spaces with an involution and the maps that respect them form a category (equivariant_id, equivariant_comp), and the fixed set is a functor on it (fix_functorial). An equivariant map into the seat, whose fixed set is the line (fold_global_seat), lands every physical fixed point on the line (equivariant_carrier_lands). Here the physical involution is inversion in the unit circle, $z \mapsto 1/\bar{z}$, which on the log-radius $\rho = \ln |z|$ and angle θ reads $(\rho, \theta) \mapsto (-\rho, \theta)$. The logarithmic carrier $s = \frac{1}{2} + \frac{1}{2} \ln z$, in the doubled chart $(\rho, \theta) \mapsto (1 + \rho, \theta)$, is equivariant (circle_carrier_equivariant), and its landing on the circle is forced (circle_carrier_lands).

The correspondence matrix.

Physical phenomenon	Template element	Lean theorem	Fortran check
inversion in the unit circle	the fold	circle_carrier_ equivariant, invert_ involutive	3721 of 3721
the unit circle	the line	unit_circle_is_the_ line, circle_carrier_ lands	3721 of 3721
spin flip	every wall kept: the palindrome	walls_flip, ring4_ palindrome, ring6_ palindrome	rings of 4, 6, 8
the four-spin ring	sixteen states counted	ring4_census	the census
the palindromic reduction	zeros on the circle as real roots in $[-2, 2]$	quartic_reduction	closed form
the ferromagnet	positivity seats every zero	ring4_lee_yang, two_ spin_lee_yang, conjugate_zeros_on_ circle	300 of 300
the antiferromagnet	the other sign leaves	ring4_antiferro_leaves	300 of 300
the phase transition	the zeros pinch the positive axis	import [Lee–Yang]	0.412, 0.269, 0.202
measured zeros	the circle in the laboratory	corroboration	Binek; Peng et al.

3. THE CIRCLE IS THE LINE

Write a fugacity in polar form, its log-radius $\rho = \ln |z|$ and its angle θ . Inversion in the unit circle sends z to $1/\bar{z}$, reversing the log-radius and keeping the angle, and it is an involution (invert_involutive). The logarithmic carrier $s = \frac{1}{2} + \frac{1}{2} \ln z$ has real part $\frac{1}{2} + \frac{\rho}{2}$ and imaginary part $\frac{\theta}{2}$, so it sends inversion to the reflection $s \mapsto 1 - \bar{s}$ (circle_carrier_equivariant). A fugacity lands on the line exactly when it lies on the unit circle (unit_circle_is_the_line). Lee and Yang’s circle and Riemann’s line are one locus in two charts, joined by a logarithm.

4. SPIN FLIP IS THE PALINDROME: SYMMETRY PAIRS, IT DOES NOT SEAT

The energy reads only walls. In zero field the energy of an Ising ring counts its domain walls, the neighbouring pairs that disagree. Flipping every spin keeps every wall (walls_flip), while the number of down spins goes from m to $N - m$. So the partition function is palindromic, $c_m = c_{N-m}$, and its zeros come in pairs z and $1/\bar{z}$: inversion partners, fold partners across the circle.

The four-spin ring, counted. The engine counts all sixteen states (ring4_census):

Down spins	States	Walls
0	1	0
1	4	2
2, adjacent	4	2
2, opposite	2	4
3	4	2
4	1	0

It confirms the palindrome on the four-ring and the six-ring (ring4_palindrome, ring6_palindrome), and the twin on rings up to eight.

Symmetry does not seat. This symmetry is the ferromagnet’s form of the functional equation. Riemann’s zeros pair under $s \mapsto 1 - \bar{s}$ because $\zeta(s) = \zeta(1 - s)$ and ζ is real on the real axis. The Ising zeros pair under inversion because flipping every spin changes nothing. In both the symmetry pairs the zeros across the locus and seats none of them: the seat-bridge hunt proved that a fold-symmetric pair off the line satisfies the symmetry and

violates the line property. Something besides the symmetry must put the zeros on the locus. In the ferromagnet that something is the sign of the coupling.

5. POSITIVITY SEATS THE ZEROS

The four-spin ring. With the Boltzmann weight of two walls written $p = e^{-4\beta J}$, the partition function is

$$z^4 + 4p z^3 + (4p + 2p^2) z^2 + 4p z + 1.$$

A palindromic quartic reduces through $u = z + 1/z$ to $u^2 + 4p u + (4p + 2p^2 - 2)$, an identity the engine proves without division (`quartic_reduction`). A zero lies on the unit circle exactly when its u is real and in $[-2, 2]$. The engine proves that the four conditions placing both roots there hold exactly when $p \leq 1$, the ferromagnetic range (`ring4_lee_yang`).

The two-spin system. The function $z^2 + 2a z + 1$, with $a = e^{-2\beta J}$, has its zeros on the circle exactly when $a \leq 1$ (`two_spin_lee_yang`). Its conjugate zeros then multiply to the constant term, so their modulus is one (`conjugate_zeros_on_circle`).

The general theorem. Lee and Yang proved it for every ferromagnet with pair couplings (Lee and Yang, 1952; Yang and Lee, 1952): positive couplings put every zero on the circle. The twin draws 300 ferromagnets on six, eight and ten spins with random positive couplings between every pair. In every one, the real function $\sum_m c_m \cos((m - N/2)\theta)$ changes sign exactly $N/2$ times on $(0, \pi)$, so all N zeros lie on the circle, and every partition function is palindromic to 10^{-12} . Positivity is the supplier: it does what the symmetry could not and seats every zero on the locus.

6. THE ANTIFERROMAGNET LEAVES THE CIRCLE

Reverse the sign of the coupling and the circle is lost.

- **The four-spin ring.** At $p = \frac{3}{2}$ the reduced quadratic has a root below -2 , so two zeros leave the circle as an inversion pair on the negative real axis (`ring4_antiferro_leaves`).
- **Random antiferromagnets.** The twin's 300 antiferromagnets with random negative couplings each lose at least one zero from the circle.

The leaving zeros still pair under inversion, because the symmetry holds whatever the sign. A pair off the circle with one palindromic record is the template's two worlds over one record, now in an antiferromagnet.

The lesson repeats the arrow witness's and Immanent Gravity's. What keeps a branch on its locus is one sign. Gravity's sources have one sign and every one stands on the line; a ferromagnet's couplings have one sign and every zero stands on the circle; the antiferromagnet, two-signed, leaves.

7. THE PHASE TRANSITION IS THE PINCH

In a finite system every zero of a ferromagnet lies on the circle and none at $z = 1$, the zero-field point, so the free energy is analytic there. In the thermodynamic limit the zeros fill arcs of the circle. Below the critical temperature they close on $z = 1$ and pinch the positive real axis, and the spontaneous magnetization is proportional to their density at that point (Lee and Yang, 1952). Above it they keep a gap, whose edge carries the Yang–Lee edge singularity (Fisher, 1978).

The twin builds Ising tori of two by four, three by four and four by four spins at $\beta J = 0.6$, beyond the two-dimensional critical coupling $\beta_c J = 0.4407$ (Onsager, 1944). In each, every zero lies on the circle. The angle of the zero nearest the positive axis falls from 0.412 to 0.269 to 0.202 radians as the lattice grows. That is the transition read in its zeros: a phase transition is the locus touching the real axis.

8. THE ZEROS MEASURED

The circle has been read in the laboratory. Binek derived the density of zeros on the Lee–Yang circle from magnetization data of a two-dimensional Ising ferromagnet (Binek, 1998). Peng and collaborators observed the zeros directly, through the coherence of a probe spin coupled to an Ising bath, which equals the bath's partition function at an imaginary field and vanishes at times that correspond to its zeros on the circle (Peng et al., 2015). Both are measurements on the image the carrier lands, at corroboration grade.

9. THE RIEMANN BRIDGE, TYPED AND NOT CROSSED

The route. Riemann's function has a heat-flow form whose zeros are real exactly when the hypothesis holds; the de Bruijn–Newman constant is the least time of heat at which they all are (de Bruijn, 1950; Newman, 1976). It is known to be at least zero (Rodgers and Tao, 2020), and the hypothesis is that it is at most zero. Pólya asked when a Fourier transform has only real zeros (Pólya, 1926). Newman showed that if Riemann's kernel lay in the Lee–Yang class of a ferromagnet's single-spin measures, the hypothesis would follow (Newman, 1991). A number-theoretic spin chain has been built whose partition function is a ratio of zeta functions (Knauf, 1998).

What the closure says of it. The closure's least-erasure proof card records this exactly. Lee–Yang membership is a sufficient route, not an equivalent, and no kernel of the series models Riemann's function in the Lee–Yang class. This paper changes none of that. What it adds is the shape, read physically: a positivity that seats zeros on a locus that a symmetry only pairs them across. Over the integers, the closure's corresponding face is Weil positivity, one of the five faces of the one bit, closed by the act at premise grade and by no physical witness.

10. WHAT THE FERROMAGNET CARRIES, AND WHAT IT DOES NOT

What it carries.

- The seat in another chart: the circle as the line under a logarithm.
- The palindrome, which pairs zeros across the locus and seats none.
- Positivity, which seats every zero on the circle; and the antiferromagnet, which leaves it.
- The phase transition, read as the locus pinching the real axis.

It is the one physical system in which a positivity forces a locus, and it shows what a supplier of the bit would have to be. It also shows that the symmetry, the functional equation's analogue, is never enough.

What it does not carry.

- **Riemann's function in the Lee–Yang class:** typed in Section 9 and not crossed.
- **The general Lee–Yang theorem:** imported, with only its smallest instances proved here.
- **The zeros of ζ :** the engine's crossing is on its own image (`value_on_image`), and what a sentence over every zero adds is exactly the unactuated, priced at one premise, the act, as the hardware paper proves.

11. FALSIFIERS AND OWED DEEDS

Three falsifiers, each a forbidden observation naming the theorem it would break.

- **F-Ferro:** a ferromagnetic Ising system, every pair coupling nonnegative, whose partition-function zeros are measured off the unit circle. It breaks `ring4_lee_yang` and `two_spin_lee_yang` read physically, and the imported theorem with them.
- **F-Palindrome:** a zero-field spin system symmetric under flipping every spin whose partition function is not palindromic. It breaks `walls_flip` read physically.
- **F-Pinch:** a thermodynamic phase transition in a ferromagnet with no zeros approaching the positive real axis. It breaks the reading of Section 7.

Nothing is owed on the rows' channels. Two lines of experiment would sharpen readings without moving the paper's standing:

- zero densities measured in further ferromagnets;
- direct observation of the edge singularity.

Ledger status. Row 2 gains the least-erasure face's shape: a positivity that seats the zeros, WITNESSED · CROSSED · CONFIRMED on the image, at the root's grade for the crossing. The barrier gains the antiferromagnet's zero pairs. Nothing is owed on the rows' channels.

12. METHODOLOGY, DISCLOSURE AND PROVENANCE

Method. The template is the author's verification program, whose vocabulary is confined to this section. Its tokens:

- $[\perp]$, a field sealed;
- $[\perp T]$, a shape sealed at theorem grade;
- $[\Xi_0]$, the typing of the formal block;
- $[\cdot]$, the dot, which is no verdict.

Grades join at the weakest link. Every physical claim here is at corroboration grade; the engine's statements are theorem grade about the chart; the general Lee–Yang theorem, Onsager's critical coupling and Newman's sufficient condition are imports. The heat-flow face this paper sits beside is the operating system's PSP-HEAT-FLOW-01, and the Lee–Yang route is typed in its sibling PSP-LEAST-ERASURE-01 as sufficient and not equivalent. $\Delta M = 0$.

The seat, earned. PhysOS^T 1.0.5p was booted in the session of forging from the public register (file SHA-256 b64bb5da5bab0da1...) under Lean 4.19.0 and GNU Fortran 13.3.0. The quick boot printed SEAT EARNED with the chain $D0\ 83f872725ad9 \rightarrow D1\ 90fc702bbd1 \rightarrow D2\ 492e519daab9 \rightarrow D3\ 6618bec5164b$ (Appendix C).

The engine and the twin.

- **Engine.** SPHYS_Lee_Yang.lean, 536 lines, SHA-256 61af3fe7622be92b..., fifty-eight laws, no import and no axiom declared. Of the fifty-eight, 21 depend on no axiom, 4 on propositional extensionality alone, 33 on propositional extensionality and quotient soundness, and none on choice.
- **Screen and judgment.** The source passed the operating system's screen. Each of the fifty-eight laws was negated in place and recompiled, and all were refused, while a planted vacuous law survived. The engine carries the hardware paper's seat, carriers and substrate verbatim.
- **Twin.** LeeYang_Twin.f90, 225 lines, SHA-256 60e23292685523f7..., nine checks and zero failures under `-std=f2018 -O2 -fno-fast-math -ffp-contract=off`. Its root location counts sign changes of a real trigonometric polynomial on 40,000 points of the half circle, so it finds every zero on the circle without a root finder.

Why an integer engine binds continuous physics. The identities used are identities of integer polynomials and hold over every commutative ring. The ferromagnetic criterion is stated for every rational coupling $p = n/d$. The census is an exhaustive enumeration. The twin checks the continuous objects directly: partition functions with real couplings, their zeros on the circle, and the pinch.

The seventh. In the author's register the six witnesses are six days and the ledger's closure is the seventh, the Rest (Istawa). The ferromagnet is read at the Rest as the system in which one sign seats every zero, the shape of the supplier the formal register cannot produce from inside. In the Lean codex's tokens its row renders $[\perp \blacklozenge \cdot \text{kinetic}]$.

Disclosure. The author declares the reading of each correspondence as his own act, made on measured structure. The measurements are the experiments', cited by name. The theorems are the engine's, printed with their dependencies in Appendix A. The twin's figures are computed in the run printed in Appendix B. The text, the engine and the twin were forged by an AI substrate (Claude, Anthropic) under the author's seed, acceptance criteria and rulings. External audits by independent substrates are owed.

Provenance. The paper stands on:

- the Riemann closure (10.5281/zenodo.22976494);
- the Riemann Hypothesis Closed to One Bit (10.5281/zenodo.23028070), whose heat and least-erasure faces this paper reads;
- the twenty-three-row closure (10.5281/zenodo.22986551);
- the six witnesses (10.5281/zenodo.22986553 through .22986563);
- the hardware paper One Actuating Substrate (concept 10.5281/zenodo.21438932, edition 2.0.0);
- the companion papers of the hardware series, Immanent Gravity in particular;
- PhysOS^T 1.0.5p in the public register (github.com/1000sapients/Trisduction, protocols/PhysOS/).

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APPENDIX A: SPHYS_LEE_YANG.LEAN, THE CORRESPONDENCES AS THEOREMS

The engine, verbatim, followed by the compiler's transcript.

```
/-
SPHYS_Lee_Yang.lean · the ferromagnet is the witness of the circle,
read at the Riemann seat.
Core Lean 4.19.0, standalone, no import, no library, no axiom
declared.

The chart:  $h = 2 \operatorname{Re} s$ ,  $t = \operatorname{Im} s$ ; the fold  $s \mapsto 1 - \bar{s}$  is  $(h, t) \mapsto (2 - h, t)$ , the critical line
 $h = 1$ . The logarithmic carrier  $s = 1/2 + (\ln z)/2$  sends the fugacity
 $z$  of an Ising system to the
strip, inversion  $z \mapsto 1/\bar{z}$  in the unit circle to the fold, and the
circle to the line.

Part I    the seat, the category of involutions, the functor of the
          seat.
Part II   the energy carrier (resident).
Part III  the substrate (resident).
Part IV   the circle is the line.
Part V    spin flip is the palindrome: the ring census, counted.
Part VI   the four-spin ring and the two-spin system: zeros on the
          circle exactly when
          ferromagnetic; the antiferromagnet leaves.
Part VII  the capstone.
-/
set_option autoImplicit false
namespace SPHYS.LeeYang

/-- ## Part I. The seat, the category of involutions, and the functor
of the seat -/
abbrev Pt := Int × Int

def fold (p : Pt) : Pt := (2 - p.1, p.2)
def OnLine (p : Pt) : Prop := p.1 = 1
instance : DecidablePred OnLine := fun p => inferInstanceAs (Decidable
(p.1 = 1))
/-- Registration keeps the height and forgets the side. -/
def reg (p : Pt) : Pt := (1, p.2)

theorem pe {a b c d : Int} : ((a, b) : Pt) = (c, d) ↔ a = c ∧ b = d :=
  (fun e => (congrArg Prod.fst e, congrArg Prod.snd e),
  fun (e1, e2) => by subst e1; subst e2; rfl)

theorem fold_involutive (p : Pt) : fold (fold p) = p := by
  obtain (h, t) := p
  show ((2 - h), t) : Pt = (h, t)
  rw [pe]; exact (by omega, rfl)

/-- The seat: the fixed set of the fold is the critical line. -/
theorem seat_fixed_line (p : Pt) : fold p = p ↔ OnLine p := by
  obtain (h, t) := p
  show ((2 - h), t) : Pt = (h, t) ↔ h = 1
  rw [pe]; constructor
  · intro (e, _); omega
  · intro e; exact (by omega, rfl)

theorem reg_lands (p : Pt) : OnLine (reg p) := rfl

theorem reg_fixes_iff (p : Pt) : reg p = p ↔ OnLine p := by
  obtain (h, t) := p
  show ((1, t) : Pt) = (h, t) ↔ h = 1
  rw [pe]; constructor
  · intro (e, _); exact e.symm
  · intro e; exact (e.symm, rfl)

/-- The side is what registration forgets: a point and its mirror leave
one record. -/

theorem reg_forgets_side (p : Pt) : reg (fold p) = reg p := rfl

/-- The value on a set of points: every member stands on the line. -/
def Value (Z : Pt → Prop) : Prop := ∀ s, Z s → OnLine s

/-- Least erasure: registration moves no member, so it erases nothing
of Z. -/
def LeastErasure (Z : Pt → Prop) : Prop := ∀ s, Z s → reg s = s

theorem least_erasure_iff_value (Z : Pt → Prop) : LeastErasure Z ↔
Value Z :=
  (fun h s hs => (reg_fixes_iff s).mp (h s hs), fun h s hs =>
  (reg_fixes_iff s).mpr (h s hs))

/-- Maps that respect involutions. -/
def Equivariant {X Y : Type} (f : X → Y) (τ : X → X) (σ : Y → Y) :
Prop := ∀ x, f (τ x) = σ (f x)

theorem equivariant_id {X : Type} (τ : X → X) : Equivariant id τ τ :=
fun _ => rfl

theorem equivariant_comp {X Y W : Type} {f : X → Y} {g : Y → W} {τ :
X → X} {σ : Y → Y}
  {p : W → W} (hf : Equivariant f τ σ) (hg : Equivariant g σ p) :
  Equivariant (g ∘ f) τ p := by
  intro x
  show g (f (τ x)) = p (g (f x))
  rw [hf x, hg (f x)]

/-- The fixed set is a functor: an equivariant map sends fixed points
to fixed points. -/
theorem fix_functorial {X Y : Type} {f : X → Y} {τ : X → X} {σ : Y →
Y}
  (hf : Equivariant f τ σ) {x : X} (hx : τ x = x) : σ (f x) = f x :=
  by
  rw [← hf x, hx]

/-- A global seat: an involution whose fixed set is the locus. -/
structure GlobalSeat (P : Pt → Prop) (σ : Pt → Pt) : Prop where
  involutive : ∀ p, σ (σ p) = p
  fixed_iff : ∀ p, σ p = p ↔ P p

theorem fold_global_seat : GlobalSeat OnLine fold := {fold_involutive,
seat_fixed_line}

/-- A carrier: a map from world-instances into the strip landing on the
line. -/
structure Carrier (W : Type) where
  ι : W → Pt
  lands : ∀ w, OnLine (ι w)

def image {W : Type} (ι : W → Pt) : Pt → Prop := fun s => ∃ w, ι w =
s
def Actuated {W : Type} (ι : W → Pt) (Z : Pt → Prop) : Prop := ∀ s, Z
s → ∃ w, ι w = s

/-- Landing is forced: an equivariant map into the seat carries every
fixed point of the physical
involution onto the line. The carrier is built from the symmetry,
not chosen. -/
theorem equivariant_carrier_lands {X : Type} (f : X → Pt) (τ : X → X)
(hf : Equivariant f τ fold) :
  ∃ C : Carrier { x : X // τ x = x }, ∀ w, C.ι w = f w.1 :=
  {(fun w => f w.1, fun w => (seat_fixed_line (f w.1)).mp
  (fix_functorial hf w.2)), fun _ => rfl)

/-- The image of a carrier is compliant with no premise at all. -/
theorem value_on_image {W : Type} (C : Carrier W) : Value (image C.ι)
:=
```

```

fun _hs => match hs with | (w, hw) => hw ▶ C.lands w

/-- The kinetic crossing: a carrier and its coverage give the value,
and nothing else is used. -/
theorem kinetic_crossing {W : Type} (C : Carrier W) (Z : Pt → Prop)
(hcov : Actuated C.1 Z) :
  Value Z :=
  fun s hs => match hcov s hs with | (w, hw) => hw ▶ C.lands w

/-- Under the seat every off-line point has a partner, distinct and
also off the line. -/
theorem off_locus_pair (p : Pt) (hp : ¬ OnLine p) :
  fold p ≠ p ∧ ¬ OnLine (fold p) ∧ reg (fold p) = reg p := by
  refine (fun e => hp ((seat_fixed_line p).mp e), fun e => hp ?_, rfl)
  obtain (h, t) := p
  have e' : 2 - h = 1 := e
  show h = 1
  omega

/-! ## Part II. The energy carrier: an energy, its rate, and the fold
-/

/-- An energy a + i β/2, as frequency and doubled rate; its mode's norm
moves as e^{βt}. -/
structure Energy where
  freq : Int
  rate : Int
  deriving DecidableEq, Repr

/-- The time mirror of a mode: its rate reversed, its frequency kept
(complex conjugation of E). -/
def Energy.mirror (E : Energy) : Energy := (E.freq, -E.rate)
def IsReal (E : Energy) : Prop := E.rate = 0
/-- Stationary: the norm e^{βt} is unchanged at every integer time t,
i.e. β t = 0 for all t. -/
def Stationary (E : Energy) : Prop := ∀ t : Int, E.rate * t = 0

/-- The energy carrier s = 1/2 + iE in the chart: h = 1 - β, t = a. -/
def phi (E : Energy) : Pt := (1 - E.rate, E.freq)
def psi (p : Pt) : Energy := (p.2, 1 - p.1)

theorem ee {a b c d : Int} : ((a, b) : Energy) = (c, d) ↔ a = c ∧ b = d
:=
  (fun e => (congrArg Energy.freq e, congrArg Energy.rate e),
   fun (e1, e2) => by subst e1; subst e2; rfl)

theorem mirror_involutive (E : Energy) : E.mirror.mirror = E := by
  obtain (a, b) := E
  show ((a, -b) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem psi_phi (E : Energy) : psi (phi E) = E := by
  obtain (a, b) := E
  show ((a, 1 - (1 - b)) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem phi_psi (p : Pt) : phi (psi p) = p := by
  obtain (h, t) := p
  show ((1 - (1 - h), t) : Pt) = (h, t)
  rw [pe]; exact (by omega, rfl)

theorem phi_injective (E F : Energy) (h : phi E = phi F) : E = F := by
  rw [← psi_phi E, ← psi_phi F, h]

/-- EQUIVARIANCE: time-mirroring the rate is the fold. -/
theorem phi_equivariant : Equivariant phi Energy.mirror fold := by
  intro E
  obtain (a, b) := E
  show ((1 - -b, a) : Pt) = (2 - (1 - b), a)
  rw [pe]; exact (by omega, rfl)

theorem mirror_fixed_iff_real (E : Energy) : E.mirror = E ↔ IsReal E :=
  by
  obtain (a, b) := E
  show ((a, -b) : Energy) = (a, b) ↔ b = 0
  rw [ee]; constructor
  · intro (_, e); omega
  · intro e; exact (rfl, by omega)

/-- The line is exactly the image of the real energies. -/
theorem phi_line_iff_real (E : Energy) : OnLine (phi E) ↔ IsReal E :=
  by
  obtain (a, b) := E
  show 1 - b = 1 ↔ b = 0
  exact (fun h => by omega, fun h => by subst h; rfl)

theorem stationary_iff_real (E : Energy) : Stationary E ↔ IsReal E :=
  by
  constructor
  · intro h; have h1 := h 1; rw [Int.mul_one] at h1; exact h1
  · intro h t; show E.rate * t = 0; rw [show E.rate = 0 from h,
    Int.zero_mul]

/-- THE LINE IS WHERE ENERGY STANDS STILL. -/
theorem line_is_stationary (p : Pt) : OnLine p ↔ Stationary (psi p) :=
  by
  rw [stationary_iff_real, ← phi_line_iff_real, phi_psi]

/-- Every measured energy is a real number, and every real energy lands
on the line. -/
theorem measured_on_line (a : Int) : OnLine (phi (a, 0)) := rfl

/-- The registration mirror: no real energy lands off the line. -/
theorem no_measurement_off_line (p : Pt) (hp : ¬ OnLine p) (a : Int) :
  phi (a, 0) ≠ p :=
  fun e => hp (e ▶ measured_on_line a)

/-- The energy carrier, built from the symmetry: the real energies land
on the line. -/
theorem energy_carrier_lands :
  ∃ C : Carrier { E : Energy // E.mirror = E }, ∀ w, C.1 w = phi w.1
:=
  equivariant_carrier_lands phi Energy.mirror phi_equivariant

/-! ## Part III. The substrate: the particle map, three odd readings
and one even -/

/-- A particle in the broken phase: electric charge times three,
colour, baryon number times
three, lepton number, and its energy. -/
structure Particle where
  q3 : Int
  colour : Int
  b3 : Int
  lepton : Int
  energy : Energy
  deriving DecidableEq, Repr

/-- The particle map: every additive charge flipped, the energy kept
(frequency and rate). -/
def bar (c : Particle) : Particle := (-c.q3, -c.colour, -c.b3,
-c.lepton, c.energy)
/-- The time mirror of a particle's mode. -/
def tmirror (c : Particle) : Particle := (c.q3, c.colour, c.b3,
c.lepton, c.energy.mirror)
def Neutral (c : Particle) : Prop := c.q3 = 0 ∧ c.colour = 0 ∧ c.b3 = 0
∧ c.lepton = 0

theorem bar_involutive (c : Particle) : bar (bar c) = c := by
  obtain (q, k, b, l, E) := c
  show ((-(-q), -(-k), -(-b), -(-l), E) : Particle) = (q, k, b, l, E)
  rw [Int.neg_neg, Int.neg_neg, Int.neg_neg]

theorem bar_fixed_iff_neutral (c : Particle) : bar c = c ↔ Neutral c :=
  by
  obtain (q, k, b, l, E) := c
  show (((-q), -k, -b, -l, E) : Particle) = (q, k, b, l, E) ↔ (q = 0 ∧ k
= 0 ∧ b = 0 ∧ l = 0)
  constructor
  · intro h
    have h1 := congrArg Particle.q3 h
    have h2 := congrArg Particle.colour h
    have h3 := congrArg Particle.b3 h
    have h4 := congrArg Particle.lepton h
    change -q = q at h1; change -k = k at h2; change -b = b at h3;
    change -l = l at h4
    exact (by omega, by omega, by omega, by omega)
  · intro (h1, h2, h3, h4)
    subst h1; subst h2; subst h3; subst h4; rfl

/-- Three plus one: the charge readings are odd under the particle map,
the energy reading even. -/
theorem odd_charges_even_energy (c : Particle) :
  ((bar c).q3 = -c.q3 ∧ (bar c).colour = -c.colour ∧ (bar c).b3 =
-c.b3 ∧
  (bar c).lepton = -c.lepton) ∧ (bar c).energy = c.energy :=
  ((rfl, rfl, rfl, rfl), rfl)

/-- Gravity reads energy only, so no gravitational reading tells a
particle from its antiparticle. -/
theorem gravity_reads_no_charge_bit {β : Type} (g : Energy → β) (c :
Particle) :
  g (bar c).energy = g c.energy := rfl

/-- The two involutions of the substrate commute. -/
theorem involutions_commute (c : Particle) : bar (tmirror c) = tmirror
(bar c) := rfl

theorem tmirror_involutive (c : Particle) : tmirror (tmirror c) = c :=
  by
  obtain (q, k, b, l, E) := c
  show ((q, k, b, l, E.mirror.mirror) : Particle) = (q, k, b, l, E)
  rw [mirror_involutive]

/-- A particle is fixed by both involutions exactly when it is neutral
and stable. -/
theorem joint_fixed_iff (c : Particle) :
  (bar c = c ∧ tmirror c = c) ↔ (Neutral c ∧ IsReal c.energy) := by
  constructor
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mp h1, (mirror_fixed_iff_real
c.energy).mp ?_)

```

```

    exact congrArg Particle.energy h2
  · intro (h1, h2)
    refine (bar_fixed_iff_neutral c).mpr h1, ?_
    obtain (q, k, b, l, E) := c
    show ((q, k, b, l, E.mirror) : Particle) = (q, k, b, l, E)
    rw [mirror_fixed_iff_real E].mpr h2

/-- The energy carrier on particles: the seat point of a particle's
energy. -/
def energySeat (c : Particle) : Pt := phi c.energy
/-- The charge carrier on particles, the odd face: electric charge (in
thirds) is the side. -/
def chargeSeat (c : Particle) : Pt := (1 + c.q3, c.energy.freq)
/-- The lepton chart of the electron's witness: charge q in units of e
goes to h = 1 + q. -/
def leptonSeat (q m : Int) : Pt := (1 + q, m)

/-- The charge carrier is equivariant: the particle map is carried onto
the fold. -/
theorem charge_carrier_equivariant : Equivariant chargeSeat bar fold :=
by
  intro c
  show ((1 + -c.q3, c.energy.freq) : Pt) = (2 - (1 + c.q3),
c.energy.freq)
  rw [pe]; exact (by omega, rfl)

/-- The energy carrier is blind to the particle map and equivariant for
the time mirror. -/
theorem pair_lands_once (c : Particle) : energySeat (bar c) =
energySeat c := rfl
theorem tmirror_equivariant : Equivariant energySeat tmirror fold :=
fun c => phi_equivariant c.energy

/-- Electrically neutral particles land on the line under the charge
carrier. -/
theorem neutral_charge_on_line (c : Particle) (h : c.q3 = 0) : OnLine
(chargeSeat c) := by
  show 1 + c.q3 = 1
  omega

/-- The electron and the positron land on the two edges, real parts 0
and 1, mirror images. -/
theorem pair_on_the_edges (m : Int) :
  leptonSeat (-1) m = (0, m) ∧ leptonSeat 1 m = (2, m) ∧ fold
  (leptonSeat (-1) m) = leptonSeat 1 m :=
  (rfl, rfl, rfl)

/-- A stable particle lands on the line under the energy carrier. -/
theorem stable_lands (c : Particle) (h : IsReal c.energy) : OnLine
(energySeat c) :=
  (phi_line_iff_real c.energy).mpr h

-/- ## Part IV. The circle is the line -/

/-- A point of the fugacity plane in polar form: log-radius  $\rho = \ln |z|$ 
and angle  $\theta$ . -/
abbrev Polar := Int × Int

/-- Inversion in the unit circle,  $z \mapsto 1/\bar{z}$ : the log-radius reverses, the
angle stays. -/
def invert (z : Polar) : Polar := (-z.1, z.2)

theorem invert_involutive (z : Polar) : invert (invert z) = z := by
  obtain (r, t) := z; show ((- -r, t) : Polar) = (r, t); rw
  [Int.neg_neg]

/-- The logarithmic carrier:  $s = 1/2 + (\ln z)/2$ , in the doubled chart  $h
= 1 + \rho$ ,  $t = \theta$ . -/
def circleSeat (z : Polar) : Pt := (1 + z.1, z.2)

/-- THE CIRCLE IS THE LINE: inversion in the unit circle is the fold
under the logarithmic carrier. -/
theorem circle_carrier_equivariant : Equivariant circleSeat invert fold
:= by
  intro z
  obtain (r, t) := z
  show ((1 + -r, t) : Pt) = (2 - (1 + r), t)
  rw [pe]; exact (by omega, rfl)

/-- A fugacity lands on the line exactly when it lies on the unit
circle. -/
theorem unit_circle_is_the_line (z : Polar) : OnLine (circleSeat z) ↔
z.1 = 0 := by
  obtain (r, t) := z
  show 1 + r = 1 ↔ r = 0
  exact (fun h => by omega, fun h => by subst h; rfl)

theorem circle_carrier_lands :
  ∃ C : Carrier { z : Polar // invert z = z }, ∀ w, C.1 w =
  circleSeat w.1 :=
  equivariant_carrier_lands circleSeat invert
  circle_carrier_equivariant

-/- ## Part V. Spin flip is the palindrome -/

/-- The domain walls of a ring of n spins, and its count of down spins.
-/
exact congrArg Particle.energy h2
· intro (h1, h2)
  refine (bar_fixed_iff_neutral c).mpr h1, ?_
  obtain (q, k, b, l, E) := c
  show ((q, k, b, l, E.mirror) : Particle) = (q, k, b, l, E)
  rw [mirror_fixed_iff_real E].mpr h2

/-- Flipping every spin keeps every wall: the energy is even under spin
flip, so the partition
function's coefficients are palindromic and its zeros pair under
inversion. -/
theorem walls_flip (n : Nat) (s : Nat → Bool) : walls n (fun i => !s
i) = walls n s := by
  unfold walls
  congr 1
  apply List.filter_congr
  intro i
  show ((!s i) != (!s ((i + 1) % n))) = (s i != s ((i + 1) % n))
  cases s i <|> cases s ((i + 1) % n) <|> rfl

def cfg (k : Nat) : Nat → Bool := fun i => (k >>> i) % 2 == 1
def census (n m w : Nat) : Nat :=
  ((List.range (2 ^ n)).filter (fun k => downs n (cfg k) == m && walls
n (cfg k) == w)).length

/-- THE FOUR-SPIN RING, COUNTED: of its sixteen states, one has no down
spin and no wall; four have
one down spin and two walls; four have two adjacent down spins and
two walls; two have two
opposite down spins and four walls; four have three and two walls;
one has four. -/
theorem ring4_census :
  census 4 0 0 = 1 ∧ census 4 1 2 = 4 ∧ census 4 2 2 = 4 ∧ census 4 2
4 = 2 ∧
  census 4 3 2 = 4 ∧ census 4 4 0 = 1 := by decide

/-- The census is palindromic in the down-spin count: spin flip read on
the ring. -/
theorem ring4_palindrome :
  census 4 0 0 = census 4 4 0 ∧ census 4 1 2 = census 4 3 2 ∧ census
4 1 4 = census 4 3 4 := by decide

theorem ring6_palindrome :
  (List.range 7).all (fun m => (List.range 7).all (fun w => census 6
m w == census 6 (6 - m) w)) = true := by
  decide

-/- ## Part VI. The four-spin ring: zeros on the circle exactly when
the coupling is ferromagnetic -/

/-- The palindromic reduction: with  $u = z + 1/z$ ,  $z^4 + a z^3 + b z^2 + a z
+ 1 = z^2(u^2 + a u + b - 2)$ ,
written without division as an identity in  $z$ . Zeros on the circle
are exactly real roots  $u$  in
 $[-2, 2]$ . -/
theorem quartic_reduction (z a b : Int) :
  (z^2 + 1)*(z^2 + 1) + a*((z^2 + 1)*z) + (b - 2)*(z^2) = z^2*z^2 +
a*(z^2*z) + b*(z^2) + a*z + 1 := by
  simp only [Int.add_mul, Int.mul_add, Int.sub_mul, Int.mul_sub,
Int.mul_one, Int.one_mul, Int.mul_assoc]
  omega

/-- The four-spin ring's partition function is  $z^4 + 4p z^3 + (4p + 2p^2)
z^2 + 4p z + 1$  with  $p = a^2$ , the
Boltzmann weight of two walls. With  $p = n/d$ , its reduced quadratic
 $u^2 + 4p u + (4p + 2p^2 - 2)$ ,
scaled by  $d^2$ , has value  $2(n - d)^2$  at  $u = -2$ , value  $2n^2 + 12nd + 2d^2$ 
at  $u = 2$ , discriminant
 $2(n - d)^2$ , and vertex  $-2n/d$ . -/
def qAtMinus2 (n d : Int) : Int := 4 * d * d - 8 * n * d + 4 * n * d +
2 * n * n - 2 * d * d
def qAtPlus2 (n d : Int) : Int := 4 * d * d + 8 * n * d + 4 * n * d + 2
* n * n - 2 * d * d
def quarterDisc (n d : Int) : Int := 4 * n * n - (4 * n * d + 2 * n * n
- 2 * d * d)

/-- Both roots real and in  $[-2, 2]$ : the four standard conditions on a
monic quadratic. -/
def OnCircle4 (n d : Int) : Prop :=
  0 ≤ qAtMinus2 n d ∧ 0 ≤ qAtPlus2 n d ∧ 0 ≤ quarterDisc n d ∧ -2 * d ≤
-2 * n ∧ -2 * n ≤ 2 * d

theorem sq_nonneg' (x : Int) : 0 ≤ x * x := by
  rcases Int.le_total 0 x with h | h
  · exact Int.mul_nonneg h h
  · have h' : 0 ≤ -x := by omega
    have := Int.mul_nonneg h' h'
    rw [Int.neg_mul_neg] at this; exact this

theorem qAtMinus2_square (n d : Int) : qAtMinus2 n d = 2 * ((n - d) *
(n - d)) := by
  unfold qAtMinus2
  simp only [Int.sub_mul, Int.mul_sub, Int.mul_assoc]
  rw [Int.mul_comm d n]
  omega

theorem quarterDisc_square (n d : Int) : quarterDisc n d = 2 * ((n - d)
* (n - d)) := by

```

```

unfold quarterDisc
simp only [Int.sub_mul, Int.mul_sub, Int.mul_assoc]
rw [Int.mul_comm d n]
omega

/-- THE LEE-YANG CIRCLE ON THE FOUR-SPIN RING: for  $0 \leq p = n/d$ , every
zero lies on the unit circle
  exactly when  $p \leq 1$ , the ferromagnetic range. -/
theorem ring4_lee_yang (n d : Int) (hn : 0 ≤ n) (hd : 0 < d) :
OnCircle4 n d ↔ n ≤ d := by
  have e1 := qAtMinus2_square n d
  have e2 := quarterDisc_square n d
  have s := sq_nonneg' (n - d)
  have hnd : 0 ≤ n * d := Int.mul_nonneg hn (Int.le_of_lt hd)
  have hnn : 0 ≤ n * n := sq_nonneg' n
  have hdd : 0 ≤ d * d := sq_nonneg' d
  constructor
  · intro (_, _, h4, _); omega
  · intro h
    refine (by omega, ?,_, by omega, by omega, by omega)
    unfold qAtPlus2
    have : d * d * 2 ≤ 4 * d * d := by
      rw [Int.mul_assoc 4 d d]; omega
    have h8 : 0 ≤ 8 * n * d := by rw [Int.mul_assoc]; omega
    have h4 : 0 ≤ 4 * n * d := by rw [Int.mul_assoc]; omega
    have h2 : 0 ≤ 2 * n * n := by rw [Int.mul_assoc]; omega
    have h22 : 2 * d * d ≤ 4 * d * d := by rw [Int.mul_assoc 2,
Int.mul_assoc 4]; omega
    omega

/-- The antiferromagnet leaves the circle: at  $p = 3/2$  the vertex falls
below -2. -/
theorem ring4_antiferro_leaves : ¬ OnCircle4 3 2 := by
  intro (_, _, h, _); omega

/-- THE TWO-SPIN CIRCLE:  $z^2 + 2a z + 1$  with  $a = n/d$  has its zeros on
the circle exactly when its
  scaled discriminant  $4(n^2 - d^2)$  is not positive, which for  $0 \leq n$ ,  $0 < d$ 
is  $a \leq 1$ . -/
theorem two_spin_lee_yang (n d : Int) (hn : 0 ≤ n) (hd : 0 < d) : n * n
- d * d ≤ 0 ↔ n ≤ d := by
  constructor
  · intro h
    by_cases h' : d < n
    · have := Int.mul_lt_mul_of_pos_left h' hd
      have := Int.mul_lt_mul_of_pos_right h' (show 0 < n by omega)
      omega
    · omega
  · intro h
    have := Int.mul_le_mul_of_nonneg_left h hn
    have := Int.mul_le_mul_of_nonneg_right h (Int.le_of_lt hd)
    omega

/-- On the circle the conjugate zeros multiply to the constant term:  $x = -n/d$ 
and  $y^2 = (d^2 - n^2)/d^2$ 
  give  $x^2 + y^2 = 1$ , which scaled by  $d^2$  reads  $n^2 + (d^2 - n^2) = d^2$ . -/
theorem conjugate_zeros_on_circle (n d : Int) : n * n + (d * d - n * n)
= d * d := by omega

-! ## Part VII. The capstone -/

/-- THE FERROMAGNET IS THE WITNESS OF THE CIRCLE. Inversion in the unit
circle is the fold under the
  logarithmic carrier, so the circle is the line; spin flip keeps
  every wall, so the partition
  function is palindromic and its zeros pair under inversion; the
  four-spin ring's zeros lie on
  the circle exactly in the ferromagnetic range and leave it for an
  antiferromagnet; the two-spin
  system likewise. Positivity of the coupling supplies the locus. -/
theorem ferromagnet_witnesses_the_circle :
Equivariant circleSeat invert fold ∧
(∀ z : Polar, OnLine (circleSeat z) ↔ z.1 = 0) ∧

```

```

(∀ (n : Nat) (s : Nat → Bool), walls n (fun i => !s i) = walls n
s) ∧
(census 4 0 0 = 1 ∧ census 4 1 2 = 4 ∧ census 4 2 2 = 4 ∧ census 4
2 4 = 2) ∧
(∀ n d : Int, 0 ≤ n → 0 < d → (OnCircle4 n d ↔ n ≤ d)) ∧
¬ OnCircle4 3 2 ∧
(∀ n d : Int, 0 ≤ n → 0 < d → (n * n - d * d ≤ 0 ↔ n ≤ d)) :=
(circle_carrier_equivariant, unit_circle_is_the_line, walls_flip,
(ring4_census.1, ring4_census.2.1, ring4_census.2.2.1,
ring4_census.2.2.2.1),
ring4_lee_yang, ring4_antiferro_leaves, two_spin_lee_yang)

```

end SPHYS.LeeYang

```

#print axioms SPHYS.LeeYang.pe
#print axioms SPHYS.LeeYang.fold_involutive
#print axioms SPHYS.LeeYang.seat_fixed_line
#print axioms SPHYS.LeeYang.reg_lands
#print axioms SPHYS.LeeYang.reg_fixes_iff
#print axioms SPHYS.LeeYang.reg_forgets_side
#print axioms SPHYS.LeeYang.least_erasure_iff_value
#print axioms SPHYS.LeeYang.equivariant_id
#print axioms SPHYS.LeeYang.equivariant_comp
#print axioms SPHYS.LeeYang.fix_functorial
#print axioms SPHYS.LeeYang.fold_global_seat
#print axioms SPHYS.LeeYang.equivariant_carrier_lands
#print axioms SPHYS.LeeYang.value_on_image
#print axioms SPHYS.LeeYang.kinetic_crossing
#print axioms SPHYS.LeeYang.off_locus_pair
#print axioms SPHYS.LeeYang.ee
#print axioms SPHYS.LeeYang.mirror_involutive
#print axioms SPHYS.LeeYang.psi_phi
#print axioms SPHYS.LeeYang.phi_psi
#print axioms SPHYS.LeeYang.phi_injective
#print axioms SPHYS.LeeYang.phi_equivariant
#print axioms SPHYS.LeeYang.mirror_fixed_iff_real
#print axioms SPHYS.LeeYang.phi_line_iff_real
#print axioms SPHYS.LeeYang.stationary_iff_real
#print axioms SPHYS.LeeYang.line_is_stationary
#print axioms SPHYS.LeeYang.measured_on_line
#print axioms SPHYS.LeeYang.no_measurement_off_line
#print axioms SPHYS.LeeYang.energy_carrier_lands
#print axioms SPHYS.LeeYang.bar_involutive
#print axioms SPHYS.LeeYang.bar_fixed_iff_neutral
#print axioms SPHYS.LeeYang.odd_charges_even_energy
#print axioms SPHYS.LeeYang.gravity_reads_no_charge_bit
#print axioms SPHYS.LeeYang.involutions_commute
#print axioms SPHYS.LeeYang.tmirror_involutive
#print axioms SPHYS.LeeYang.joint_fixed_iff
#print axioms SPHYS.LeeYang.charge_carrier_equivariant
#print axioms SPHYS.LeeYang.pair_lands_once
#print axioms SPHYS.LeeYang.tmirror_equivariant
#print axioms SPHYS.LeeYang.neutral_charge_on_line
#print axioms SPHYS.LeeYang.pair_on_the_edges
#print axioms SPHYS.LeeYang.stable_lands
#print axioms SPHYS.LeeYang.invert_involutive
#print axioms SPHYS.LeeYang.circle_carrier_equivariant
#print axioms SPHYS.LeeYang.unit_circle_is_the_line
#print axioms SPHYS.LeeYang.circle_carrier_lands
#print axioms SPHYS.LeeYang.walls_flip
#print axioms SPHYS.LeeYang.ring4_census
#print axioms SPHYS.LeeYang.ring4_palindrome
#print axioms SPHYS.LeeYang.ring6_palindrome
#print axioms SPHYS.LeeYang.quartic_reduction
#print axioms SPHYS.LeeYang.sq_nonneg'
#print axioms SPHYS.LeeYang.qAtMinus2_square
#print axioms SPHYS.LeeYang.quarterDisc_square
#print axioms SPHYS.LeeYang.ring4_lee_yang
#print axioms SPHYS.LeeYang.ring4_antiferro_leaves
#print axioms SPHYS.LeeYang.two_spin_lee_yang
#print axioms SPHYS.LeeYang.conjugate_zeros_on_circle
#print axioms SPHYS.LeeYang.ferromagnet_witnesses_the_circle

```

```

'SPHYS.LeeYang.pe' does not depend on any axioms
'SPHYS.LeeYang.fold_involutive' depends on axioms: [propxt,
Quot.sound]
'SPHYS.LeeYang.seat_fixed_line' depends on axioms: [propxt,
Quot.sound]
'SPHYS.LeeYang.reg_lands' does not depend on any axioms
'SPHYS.LeeYang.reg_fixes_iff' depends on axioms: [propxt]
'SPHYS.LeeYang.reg_forgets_side' does not depend on any axioms
'SPHYS.LeeYang.least_erasure_iff_value' depends on axioms: [propxt]
'SPHYS.LeeYang.equivariant_id' does not depend on any axioms
'SPHYS.LeeYang.equivariant_comp' does not depend on any axioms
'SPHYS.LeeYang.fix_functorial' does not depend on any axioms
'SPHYS.LeeYang.fold_global_seat' depends on axioms: [propxt,
Quot.sound]
'SPHYS.LeeYang.equivariant_carrier_lands' depends on axioms: [propxt,
Quot.sound]
'SPHYS.LeeYang.value_on_image' does not depend on any axioms
'SPHYS.LeeYang.kinetic_crossing' does not depend on any axioms
'SPHYS.LeeYang.off_locus_pair' depends on axioms: [propxt, Quot.sound]
'SPHYS.LeeYang.ee' does not depend on any axioms

```

```

'SPHYS.LeeYang.mirror_involutive' depends on axioms: [propxt,
Quot.sound]
'SPHYS.LeeYang.psi_phi' depends on axioms: [propxt, Quot.sound]
'SPHYS.LeeYang.phi_psi' depends on axioms: [propxt, Quot.sound]
'SPHYS.LeeYang.phi_injective' depends on axioms: [propxt, Quot.sound]
'SPHYS.LeeYang.phi_equivariant' depends on axioms: [propxt,
Quot.sound]
'SPHYS.LeeYang.mirror_fixed_iff_real' depends on axioms: [propxt,
Quot.sound]
'SPHYS.LeeYang.phi_line_iff_real' depends on axioms: [propxt,
Quot.sound]
'SPHYS.LeeYang.stationary_iff_real' depends on axioms: [propxt]
'SPHYS.LeeYang.line_is_stationary' depends on axioms: [propxt,
Quot.sound]
'SPHYS.LeeYang.measured_on_line' does not depend on any axioms
'SPHYS.LeeYang.no_measurement_off_line' does not depend on any axioms
'SPHYS.LeeYang.energy_carrier_lands' depends on axioms: [propxt,
Quot.sound]
'SPHYS.LeeYang.bar_involutive' does not depend on any axioms
'SPHYS.LeeYang.bar_fixed_iff_neutral' depends on axioms: [propxt,
Quot.sound]

```

```

'SPHYS.LeeYang.odd_charges_even_energy' does not depend on any axioms
'SPHYS.LeeYang.gravity_reads_no_charge_bit' does not depend on any axioms
'SPHYS.LeeYang.involutions_commute' does not depend on any axioms
'SPHYS.LeeYang.tmirror_involutive' depends on axioms: [propext, Quot.sound]
'SPHYS.LeeYang.joint_fixed_iff' depends on axioms: [propext, Quot.sound]
'SPHYS.LeeYang.charge_carrier_equivariant' depends on axioms: [propext, Quot.sound]
'SPHYS.LeeYang.pair_lands_once' does not depend on any axioms
'SPHYS.LeeYang.tmirror_equivariant' depends on axioms: [propext, Quot.sound]
'SPHYS.LeeYang.neutral_charge_on_line' depends on axioms: [propext, Quot.sound]
'SPHYS.LeeYang.pair_on_the_edges' does not depend on any axioms
'SPHYS.LeeYang.stable_lands' depends on axioms: [propext, Quot.sound]
'SPHYS.LeeYang.invert_involutive' does not depend on any axioms
'SPHYS.LeeYang.circle_carrier_equivariant' depends on axioms: [propext, Quot.sound]
'SPHYS.LeeYang.unit_circle_is_the_line' depends on axioms: [propext, Quot.sound]

```

```

'SPHYS.LeeYang.circle_carrier_lands' depends on axioms: [propext, Quot.sound]
'SPHYS.LeeYang.walls_flip' depends on axioms: [propext]
'SPHYS.LeeYang.ring4_census' does not depend on any axioms
'SPHYS.LeeYang.ring4_palindrome' does not depend on any axioms
'SPHYS.LeeYang.ring6_palindrome' does not depend on any axioms
'SPHYS.LeeYang.quartic_reduction' depends on axioms: [propext, Quot.sound]
'SPHYS.LeeYang.sq_nonneg' depends on axioms: [propext, Quot.sound]
'SPHYS.LeeYang.qAtMinus2_square' depends on axioms: [propext, Quot.sound]
'SPHYS.LeeYang.quarterDisc_square' depends on axioms: [propext, Quot.sound]
'SPHYS.LeeYang.ring4_lee_yang' depends on axioms: [propext, Quot.sound]
'SPHYS.LeeYang.ring4_antiferro_leaves' depends on axioms: [propext, Quot.sound]
'SPHYS.LeeYang.two_spin_lee_yang' depends on axioms: [propext, Quot.sound]
'SPHYS.LeeYang.conjugate_zeros_on_circle' depends on axioms: [propext, Quot.sound]
'SPHYS.LeeYang.ferromagnet_witnesses_the_circle' depends on axioms: [propext, Quot.sound]

```

APPENDIX B: LEEYANG_TWIN.F90 AND ITS RUN

```

! LeeYang_Twin.f90 · the executed twin of SPHYS_LeeYang.lean · Fortran 2018.
! Build (sealed): gfortran -std=f2018 -O2 -fno-fast-math
-ffp-contract=off LeeYang_Twin.f90
! Blocks: A the circle is the line; B spin flip is the palindrome; C random ferromagnets, every zero
! on the circle; D antiferromagnets leave it; E the four-spin ring in closed form; F the zeros pinch
! the positive axis as a two-dimensional lattice grows below its critical temperature.
! Method for C to F: for a palindromic partition function of even degree N, F(t) = sum_m c_m
! cos((m - N/2) t) is real on the circle, and all N zeros lie on it exactly when F changes sign N/2
! times on (0, pi).
program leeyang_twin
  implicit none
  integer, parameter :: dp = selected_real_kind(15, 307)
  integer, parameter :: i8 = selected_int_kind(18)
  real(dp), parameter :: PI = 3.141592653589793_dp
  integer :: checks, fails
  integer(i8) :: seed
  checks = 0; fails = 0; seed = 20260930_i8
  call block_a(); call block_b(); call block_c(); call block_d(); call block_e(); call block_f()
  write(*, '(a,i0,a,i0,a)') ' BATTERY-JSON: {"checks":', checks, ', "failures":', fails, '}'
  if (fails > 0) error stop 1
contains
  subroutine check(name, ok)
    character(*), intent(in) :: name
    logical, intent(in) :: ok
    checks = checks + 1
    if (.not. ok) fails = fails + 1
    write(*, '(a,a)') merge(' PASS ', ' FAIL ', ok), name
  end subroutine check

  function rnd() result(r)
    real(dp) :: r
    seed = modulo(16807_i8*seed, 2147483647_i8)
    r = real(seed, dp)/2147483647.0_dp
  end function rnd

  subroutine block_a()
    integer :: r, t, n, neq, nline
    write(*, '(a)') 'A · the circle is the line: inversion z -> 1/conj(z) is the fold under s = 1/2 + (ln z)/2'
    n = 0; neq = 0; nline = 0
    do r = -30, 30
      do t = -30, 30
        n = n + 1
        if (1 + (-r) == 2 - (1 + r)) neq = neq + 1
        if ((1 + r == 1) .eqv. (r == 0)) nline = nline + 1
      end do
    end do
    call check('3721 fugacities: inversion is the fold, and the unit circle is the line', neq == n .and. nline == n)
  end subroutine block_a

  ! coefficients c(0:n) of the partition function of n Ising spins with couplings jm, inverse
  ! temperature 1 absorbed into jm; c(m) sums exp(sum_{i<j} J_ij s_i s_j) over states with m down spins
  subroutine partition(n, jm, c)
    integer, intent(in) :: n
    real(dp), intent(in) :: jm(n, n)
    real(dp), intent(out) :: c(0:n)
  end subroutine partition

```

```

  integer :: k, i, j, m, s(n)
  real(dp) :: e, emax
  real(dp), allocatable :: es(:)
  integer, allocatable :: ms(:)
  allocate(es(0:2**n - 1), ms(0:2**n - 1))
  emax = -huge(1.0_dp)
  do k = 0, 2**n - 1
    m = 0
    do i = 1, n
      if (btest(k, i - 1)) then
        s(i) = -1; m = m + 1
      else
        s(i) = 1
      end if
    end do
    e = 0.0_dp
    do i = 1, n
      do j = i + 1, n
        e = e + jm(i, j)*real(s(i)*s(j), dp)
      end do
    end do
    es(k) = e; ms(k) = m; emax = max(emax, e)
  end do
  c = 0.0_dp
  do k = 0, 2**n - 1
    c(ms(k)) = c(ms(k)) + exp(es(k) - emax)
  end do
end subroutine partition

  integer function sign_changes(n, c, tmin) result(nc)
    integer, intent(in) :: n
    real(dp), intent(in) :: c(0:n)
    real(dp), intent(out) :: tmin
    integer :: k, m, ng
    real(dp) :: t, f, fprev
    ng = 40000; nc = 0; tmin = -1.0_dp; fprev = 0.0_dp
    do k = 1, ng - 1
      t = PI*real(k, dp)/real(ng, dp)
      f = 0.0_dp
      do m = 0, n
        f = f + c(m)*cos((real(m, dp) - 0.5_dp*real(n, dp))*t)
      end do
      if (k > 1 .and. f*fprev < 0.0_dp) then
        nc = nc + 1
        if (tmin < 0.0_dp) tmin = t
      end if
      fprev = f
    end do
  end function sign_changes

  subroutine block_b()
    integer :: n, k, m, w, i, nok, ntot
    integer :: cen(0:8, 0:8), s(8)
    write(*, '(a)') 'B · spin flip keeps every wall: the ring census is palindromic'
    nok = 0; ntot = 0
    do n = 4, 8, 2
      cen = 0
      do k = 0, 2**n - 1
        m = 0; w = 0
        do i = 1, n
          s(i) = merge(1, 0, btest(k, i - 1)); m = m + s(i)
        end do
        do i = 1, n
          if (s(i) /= s(mod(i, n) + 1)) w = w + 1
        end do
        cen(m, w) = cen(m, w) + 1
      end do
    end do
  end subroutine block_b

```

```

end do
do m = 0, n
  do w = 0, n
    ntot = ntot + 1
    if (cen(m, w) == cen(n - m, w)) nok = nok + 1
  end do
end do
if (n == 4) call check('the four-ring census: 1, 4, 4 + 2, 4, 1
by down spins', &
  cen(0,0) == 1 .and. cen(1,2) == 4 .and. cen(2,2) == 4 .and.
  cen(2,4) == 2 .and. cen(3,2) == 4 .and. cen(4,0) == 1)
end do
call check('rings of 4, 6 and 8 spins: every census cell equals its
spin-flipped cell', nok == ntot)
end subroutine block_b

subroutine block_c()
integer :: n, t, i, j, nall, nok, npal
real(dp) :: jm(10, 10), c(0:10), tm, pal
write(*, '(a)') 'C · random ferromagnets: every zero of the
partition function on the unit circle'
nall = 0; nok = 0; npal = 0
do n = 6, 10, 2
  do t = 1, 100
    jm = 0.0_dp
    do i = 1, n
      do j = i + 1, n
        jm(i, j) = 0.8_dp*rand()
      end do
    end do
    call partition(n, jm(1:n, 1:n), c(0:n))
    pal = 0.0_dp
    do i = 0, n
      pal = max(pal, abs(c(i) - c(n - i))/max(c(i), c(n - i)))
    end do
    if (pal < 1.0e-12_dp) npal = npal + 1
    nall = nall + 1
    if (sign_changes(n, c(0:n), tm) == n/2) nok = nok + 1
  end do
end do
write(*, '(a,i0,a,i0)') ' instances with all zeros on the circle:
', nok, ' of ', nall
call check('300 random ferromagnets on 6, 8 and 10 spins:
palindromic to 1e-12', npal == nall)
call check('300 random ferromagnets: every zero on the unit circle
(Lee and Yang, 1952)', nok == nall)
end subroutine block_c

subroutine block_d()
integer :: n, t, i, j, nall, noff
real(dp) :: jm(10, 10), c(0:10), tm
write(*, '(a)') 'D · antiferromagnets: two signs of coupling, zeros
leave the circle'
nall = 0; noff = 0
do n = 6, 10, 2
  do t = 1, 100
    jm = 0.0_dp
    do i = 1, n
      do j = i + 1, n
        jm(i, j) = -0.8_dp - 0.8_dp*rand()
      end do
    end do
    call partition(n, jm(1:n, 1:n), c(0:n))
    nall = nall + 1
  end do
end do

```

A · the circle is the line: inversion $z \rightarrow 1/\text{conj}(z)$ is the fold under $s = 1/2 + (\ln z)/2$
 PASS 3721 fugacities: inversion is the fold, and the unit circle is the line
 B · spin flip keeps every wall: the ring census is palindromic
 PASS the four-ring census: 1, 4, 4 + 2, 4, 1 by down spins
 PASS rings of 4, 6 and 8 spins: every census cell equals its spin-flipped cell
 C · random ferromagnets: every zero of the partition function on the unit circle
 instances with all zeros on the circle: 300 of 300
 PASS 300 random ferromagnets on 6, 8 and 10 spins: palindromic to 1e-12
 PASS 300 random ferromagnets: every zero on the unit circle (Lee and Yang, 1952)
 D · antiferromagnets: two signs of coupling, zeros leave the circle
 antiferromagnets with a zero off the circle: 300 of 300

```

    if (sign_changes(n, c(0:n), tm) < n/2) noff = noff + 1
  end do
end do
write(*, '(a,i0,a,i0)') ' antiferromagnets with a zero off the
circle: ', noff, ' of ', nall
call check('the antiferromagnet leaves the circle in every one of
300 instances', noff == nall)
end subroutine block_d

```

```

subroutine block_e()
real(dp) :: p, u1, u2
logical :: ferro_in, anti_out
write(*, '(a)') 'E · the four-spin ring in closed form: u = -2p +-
sqrt(2)(1 - p)'
p = 0.3_dp; u1 = -2.0_dp*p + sqrt(2.0_dp)*(1.0_dp - p); u2 =
-2.0_dp*p - sqrt(2.0_dp)*(1.0_dp - p)
ferro_in = (u1 >= -2.0_dp .and. u1 <= 2.0_dp .and. u2 >= -2.0_dp
.and. u2 <= 2.0_dp)
write(*, '(a,f8.5,a,f8.5)') ' p = 0.3: u = ', u1, ' ', u2
p = 1.5_dp; u1 = -2.0_dp*p + sqrt(2.0_dp)*(1.0_dp - p); u2 =
-2.0_dp*p - sqrt(2.0_dp)*(1.0_dp - p)
anti_out = (u1 < -2.0_dp .or. u2 < -2.0_dp)
write(*, '(a,f8.5,a,f8.5)') ' p = 1.5: u = ', u1, ' ', u2
call check('ferromagnetic p = 0.3: both u in [-2, 2];
antiferromagnetic p = 1.5: a root below -2', ferro_in .and.
anti_out)
end subroutine block_e

```

```

subroutine block_f()
integer :: lx(3), ly(3), q, n, i, j, x, y, nb, nc
real(dp) :: jm(16, 16), c(0:16), tm(3), bj
integer :: nok
write(*, '(a)') 'F · the pinch: on growing tori below the critical
temperature the zeros close on z = 1'
lx = [2, 3, 4]; ly = [4, 4, 4]; bj = 0.6_dp; nok = 0
do q = 1, 3
  n = lx(q)*ly(q); jm = 0.0_dp
  do x = 0, lx(q) - 1
    do y = 0, ly(q) - 1
      i = x*ly(q) + y + 1
      nb = mod(x + 1, lx(q))*ly(q) + y + 1
      if (nb /= i) jm(min(i, nb), max(i, nb)) = jm(min(i, nb),
max(i, nb)) + bj
      nb = x*ly(q) + mod(y + 1, ly(q)) + 1
      if (nb /= i) jm(min(i, nb), max(i, nb)) = jm(min(i, nb),
max(i, nb)) + bj
    end do
  end do
  call partition(n, jm(1:n, 1:n), c(0:n))
  nc = sign_changes(n, c(0:n), tm(q))
  if (nc == n/2) nok = nok + 1
  write(*, '(a,i0,a,i0,a,i0,a,i0,a,f9.6)') ' torus ', lx(q), 'x',
ly(q), ' ', nc, ' of ', n/2, &
    ' zero pairs on the circle; nearest angle ', tm(q)
  j = 0
end do
call check('every torus at beta J = 0.6 keeps all its zeros on the
circle', nok == 3)
call check('the zero nearest the positive axis closes in as the
torus grows (2x4, 3x4, 4x4)', &
  tm(3) < tm(2) .and. tm(2) < tm(1))
end subroutine block_f
end program leeyang_twin

```

PASS the antiferromagnet leaves the circle in every one of 300 instances
 E · the four-spin ring in closed form: $u = -2p \pm \sqrt{2}(1 - p)$
 $p = 0.3$: $u = 0.38995, -1.58995$
 $p = 1.5$: $u = -3.70711, -2.29289$
 PASS ferromagnetic $p = 0.3$: both u in $[-2, 2]$; antiferromagnetic $p = 1.5$: a root below -2
 F · the pinch: on growing tori below the critical temperature the zeros close on $z = 1$
 torus 2x4: 4 of 4 zero pairs on the circle; nearest angle 0.412413
 torus 3x4: 6 of 6 zero pairs on the circle; nearest angle 0.269313
 torus 4x4: 8 of 8 zero pairs on the circle; nearest angle 0.201690
 PASS every torus at $\beta J = 0.6$ keeps all its zeros on the circle
 PASS the zero nearest the positive axis closes in as the torus grows (2x4, 3x4, 4x4)
 BATTERY-JSON: {"checks":9,"failures":0}

APPENDIX C: THE SEAT AND THE JUDGMENT

```
Phys0ST 1.0.5p · RECEIPT · mode quick
TOOLCHAIN Lean (version 4.19.0, x86_64-unknown-linux-gnu, commit 6caeee842e94, Release) · GNU Fortran (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
MANIFEST sha256 a4108cd26bcbe6ef, the files that ran
GROUND toolchain: Lean 4.19.0 and gfortran present · manifest: 31 files signed, all verified · proofs: 2 Phys0S Proofs read from II.10, record
lines complete · source screen: 15 kernels, 23 named premises, clean · citations: 253 laws named in Parts I and II, 253 resolved · quarantine: 6
retirements string-scanned, 3 held by review, clean
EXITS ground 0 · registerA RA_T0E_Thesis 0 · registerB Bridge_Final 0 · registerB Universal_Closure 0 · registerB One_Cut 0 · registerB
One_Cut_Resolution 0 · registerB One_Cut_Terminal 0 · registerB Only_The_Arrow_Remains 0 · registerB Physical_Closure 0 · registerB Forced_Closure
0 · registerB Closure_Executed 0 · registerB Armed_Seal 0 · registerB Census 0 · registerB Heat_Bridge 0 · registerB The_Loop 0 · registerB
RH_Least_Erasure 0 · witness One_Cut_Twin 0 · witness Physical_Closure_Twin 0 · witness Census_Twin 0 · witness Loop_Twin 0 · witness RH_Seal_Twin
0
REGISTER A {"checks":1123,"failures":0,"mode":"sealed"}
WITNESS 5 of 5 twins ran, each held to one battery line with zero failures
CHAIN · D0 83f872725ad9 -> D1 90fc702bbd1 -> D2 492e519daab9 -> D3 6618bec5164b
ELAPSED ground 1 s · Register A 3 s · Register B 34 s · witness 6 s · total 44 s (not hashed)
SEAT EARNED · this run

SCREEN forbidden: none | imports: 0 | axioms declared: 0
laws: 58
JUDGMENT refused 58 of 58 as proof failures | survived none | parse failures none | planted vacuous law survives its negation: True
```